# It's tinker time

Some of the gadgets that we've seen of late border on mysticism and magic. A speaker that talks to you and can give you the news or switch on your lights when you ask it to? Certainly sounds like witch craft until you figure out how it's a bunch of LEDs, directional microphones hooked to a microcontroller with a wireless protocol board that gets all its information from Amazon. We're obviously talking about an Echo device. It seems like a marvel till you realise that you can use a Raspberry Pi board hooked up to your desktop speaker set to achieve the exact same result. If only you know how to go about building your own electronics project from scratch...

Well, that's where this FastTrack comes in. We'll take you through the very basic components that are essential to building electronic circuits and we'll follow that with a primer on how you can go about using each of those components and where they are essential. Also, we're focusing on the two tools which are absolutely essential with electronics projects – multimeter and soldering iron. They're like the WD-40 and Duct Tape of hobby electronics, to put it simply.

That's for the folks who'd like to get into the nitty gritty of the project but for the ones who'd like to move to more complex tasks, there's always a dev-board for the job. In fact, the humble Arduino board is quite capable on its own and should you need a little more performance, then you can step up to a Raspberry Pi board or even more powerful alternatives that